Electronic Supplementary Material for "Why is anything conscious?"

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S1 What is in this ESM

This document is the companion Electronic Supplementary Material for the main manuscript of "Why Is Anything Conscious?". It does three things. First, it gives the precise definitions that the main text uses. Every mathematical object is defined in math and then paraphrased in plain English. Second it contains full proofs, whereas in the main text the proofs are just sketches. Third, it contains full experimental protocol details and the extra plots. The contents are in part a restatement of earlier proofs and experiments from Bennett [2025a, 2024, 2026]. The primarily novel formal contribution unique to this paper is the separation theorems, formally showing the hierarchy of selves follows from generalisation optimal learning (something claimed but not so formally backed in earlier work), and the formalisation and proof of The Psychophysical Principle of Causality. Please refer to the GitHub for an even more comprehensive list of related proofs, the experiment code and so forth Bennett [2025b].

Notation bridge to the main text. The main text writes the extension of a statement x as E_x. In this ESM, we write $\mathrm{Ext}(x)$. These are the same object. The main text defines weakness as the size of an extension. In this ESM we keep that definition. Sometimes we also report $\log_2$ weakness. That is just weakness measured in bits.

S2 Stack theory primitives used in this paper

S2.1 Environments, programs, vocabularies, and statements

Definition 1 (Environment, programs, and vocabularies)**.** *An **environment** is a nonempty set Φ of mutually exclusive states. A **program** is any set of states $p \subseteq \Phi$. Let $\mathcal{P} := 2^{\Phi}$ denote the set of all programs. A **vocabulary** is any set $\mathfrak{v} \subseteq \mathcal{P}$. Unless we say otherwise, vocabularies are finite.*

Interpretation. A state is a complete way the world could be. A program is a yes or no classification of states. It is a potential fact (we do not assume about what, it is merely a point of potential difference between states in the vein of structuralism). A vocabulary is the set of properties the body can stably express and use.

Definition 2 (Abstraction layer, statements, and truth sets)**.** *Fix a vocabulary $\mathfrak{v}$. A **statement** is any subset $\ell \subseteq \mathfrak{v}$ whose truth set is nonempty. The **truth set** of a subset $\ell \subseteq \mathfrak{v}$ is*

$$T(\ell) := \bigcap_{p \in \ell} p,$$

*with the convention $T(\emptyset) = \Phi$. The **language induced by** $\mathfrak{v}$ is*

$$L_{\mathfrak{v}} := \{\, \ell \subseteq \mathfrak{v} \mid T(\ell) \neq \emptyset \,\}.$$

Interpretation. A statement is a bundle of properties that can all be true at once. The truth set is the set of world states compatible with that bundle. The induced language is the set of all bundles that are not contradictory.

Definition 3 (Completions, extensions, and equivalence)**.** *Fix a vocabulary $\mathfrak{v}$ and its language $L_{\mathfrak{v}}$. A* ***completion*** *of a statement $x \in L_{\mathfrak{v}}$ is any statement $y \in L_{\mathfrak{v}}$ such that $x \subseteq y$. The set of all completions of x is its* ***extension***

$$\mathrm{Ext}(x) := \{\, y \in L_{\mathfrak{v}} \mid x \subseteq y \,\}.$$

Two statements $x, y \in L_{\mathfrak{v}}$ are ***extension equivalent*** *when $\mathrm{Ext}(x) = \mathrm{Ext}(y)$.*

Interpretation. A completion is what you get by adding extra compatible details. The extension is the set of all ways you could add details while staying consistent.

Definition 4 (Weakness)**.** *The* ***weakness*** *of a statement $x \in L_{\mathfrak{v}}$ is*

$$w(x) := |\mathrm{Ext}(x)|.$$

When we want a log scale we use

$$W(x) := \log_2 w(x).$$

Interpretation. Weakness counts how many ways you can keep refining without breaking consistency. Higher weakness means you have left more options open. The log scale W is just weakness measured in bits.

S2.2 Tasks, correct policies, inference, and learning

Definition 5 ($\mathfrak{v}$-task)**.** *Fix a vocabulary $\mathfrak{v}$ and language $L_{\mathfrak{v}}$. A* ***$\mathfrak{v}$-task*** *is an ordered pair $\alpha = \langle I_\alpha, O_\alpha \rangle$ where $I_\alpha \subseteq L_{\mathfrak{v}}$ is the input set and $O_\alpha \subseteq \bigcup_{i \in I_\alpha} \mathrm{Ext}(i)$ is the output set.*

Interpretation. A task says which inputs can happen and which refined statements count as acceptable outputs. It is not necessarily a function. It is a constraint.

Definition 6 (Inference and correct policies)**.** *Fix a vocabulary $\mathfrak{v}$ and a task $\alpha = \langle I_\alpha, O_\alpha \rangle$. A* ***policy*** *is any statement $\pi \in L_{\mathfrak{v}}$. A policy π is* ***correct*** *for α when*

$$\Big(\bigcup_{i \in I_\alpha} \mathrm{Ext}(i) \Big) \cap \mathrm{Ext}(\pi) = O_\alpha.$$

Let Π_α denote the set of correct policies for α. The ***inference problem*** *for α is to determine Π_α.*

Interpretation. A policy is a constraint that selects which outputs are allowed. Correct means the allowed outputs match exactly the task outputs on the part of the world the task talks about.

Definition 7 (Learning)**.** *Let α and ω be tasks in the same language. We say α is a* ***subtask*** *of ω, written $\alpha \sqsubseteq \omega$, when $I_\alpha \subseteq I_\omega$ and $O_\alpha \subseteq O_\omega$. A learning system* ***generalises*** *from α to ω when it selects a policy $\pi \in \Pi_\alpha$ that is also correct for ω, meaning $\pi \in \Pi_\omega$.*

Interpretation. A parent task contains more cases than a child task. Generalisation means a policy that fits the child also works on the parent.

S2.3 Causal identities and selves

Definition 8 (Causal identity candidates)**.** *Fix an embodied language $L_{\mathfrak{v}}$. Let $INT \subseteq L_{\mathfrak{v}}$ be a set of intervention episodes and $OBS \subseteq L_{\mathfrak{v}}$ a set of matched non intervention observations. Define the candidate set*

$$\mathcal{C}(INT, OBS) := \Big\{ c \in L_{\mathfrak{v}} \setminus \{\emptyset\} \;\Big|\; (\forall\, int \in INT)\ c \subseteq int \text{ and } (\forall\, obs \in OBS)\ c \not\subseteq obs \Big\}.$$

Each $c \in \mathcal{C}(INT, OBS)$ is a causal identity candidate for whatever is common to the interventions but not already present in the observations.

Interpretation. A causal identity is a signature in the agent's own vocabulary. It is present whenever the source intervenes. It is absent in matched background observations.

Definition 9 (w-maximised causal identity)**.** *Assume $\mathcal{C}(INT, OBS) \neq \emptyset$. A **w-maximised causal identity** is any candidate $c^\star \in \mathcal{C}(INT, OBS)$ that maximises weakness among candidates*

$$|\mathrm{Ext}(c^\star)| = \max\{|\mathrm{Ext}(c)| \mid c \in \mathcal{C}(INT, OBS)\}.$$

When $\mathfrak{v}$ is finite, the maximum exists.

Interpretation. Many signatures might separate intervention from background. W-maxing picks the one that leaves the most room for later refinement.

Definition 10 (First order self)**.** *Let an organism $\mathfrak{o}$ have embodied language $L_{\mathfrak{v}_\mathfrak{o}}$. Let $INT_\mathfrak{o} \subseteq L_{\mathfrak{v}_\mathfrak{o}}$ be idealised self intervention episodes for $\mathfrak{o}$ and let $OBS_\mathfrak{o} \subseteq L_{\mathfrak{v}_\mathfrak{o}}$ be matched non self observations. Any w-maximised causal identity*

$$\mathfrak{o}^1 \in \mathcal{C}(INT_\mathfrak{o}, OBS_\mathfrak{o})$$

*is called a **first order self** for $\mathfrak{o}$.*

Interpretation. A first order self is the weakest self tag the organism can build that still separates self caused change from external change.

Definition 11 (Chain notation for nested causal identity modelling)**.** *Suppose we have two agents $\mathfrak{a}$ (Alice) and $\mathfrak{b}$ (Bob). A causal identity $c_\mathfrak{a}^\mathfrak{b}$ denotes the internal model of $\mathfrak{b}$ constructed by $\mathfrak{a}$.*

For a chain of agents $\mathfrak{x}_1, \ldots, \mathfrak{x}_k$ we write

$$c_\mathfrak{a}^{\mathfrak{x}_1\mathfrak{x}_2\cdots\mathfrak{x}_k}$$

for Alice's nested causal model in which $\mathfrak{x}_1$ models $\mathfrak{x}_2$ models ... models $\mathfrak{x}_k$. For example, $c_\mathfrak{a}^{\mathfrak{b}\mathfrak{a}}$ denotes Alice's model of Bob's model of Alice.

Interpretation. This is just a notation for higher order theory of mind.

Definition 12 (nth order self)**.** *Fix an agent $\mathfrak{a}$. An n**th order self** of $\mathfrak{a}$ is any causal identity whose chain ends at $\mathfrak{a}$ after interleaving $\mathfrak{a}$ with $n-1$ other agent models. Formally set*

$$\mathfrak{a}^1 := c_\mathfrak{a}^\mathfrak{a}.$$

For $n \geq 2$, choose agents $\mathfrak{x}_1, \ldots, \mathfrak{x}_{n-1}$ and set

$$\mathfrak{a}^n := c_\mathfrak{a}^{\mathfrak{x}_1\mathfrak{a}\mathfrak{x}_2\mathfrak{a}\cdots\mathfrak{x}_{n-1}\mathfrak{a}}.$$

Interpretation. A second order self is a model of how others model you. A third order self is a narrative self. It is a model of how others model how you will act, and how that constrains your own future policy. This is what makes long horizon identity, commitment, and trust possible.

S2.4 Organisms, protosymbols, and affect

Definition 13 (Organism)**.** *We represent the circumstances of an organism $\mathfrak{o}$ as a tuple*

$$\mathfrak{o} = \langle \mathfrak{v}_\mathfrak{o}, \mu_\mathfrak{o}, \mathfrak{p}_\mathfrak{o}, <_\mathfrak{o} \rangle.$$

Here $\mathfrak{v}_\mathfrak{o}$ is a vocabulary. $\mu_\mathfrak{o}$ is a viability task that describes fit behaviour for $\mathfrak{o}$. $\mathfrak{p}_\mathfrak{o}$ is a set of policies the organism is assumed to know. $<_\mathfrak{o}$ is a preference ordering over tasks.

Interpretation. This is a compact way to say. The organism has an embodied language. It has a viability constraint. It has limited memory and limited skills. It has preferences.

Definition 14 (Protosymbol system)**.** *Let $\mathfrak{o}$ be an organism. The **protosymbol system** of $\mathfrak{o}$ is*

$$\mathfrak{s}_\mathfrak{o} := \{\alpha \mid \exists \pi \in \mathfrak{p}_\mathfrak{o} \ s.t. \ \pi \in \Pi_\alpha\}.$$

Interpretation. A protosymbol is a reusable meaning template. It is a task the organism already knows how to solve. The protosymbol system is the set of those templates.

Definition 15 (Affect)**.** *Let $\mathfrak{o}$ and $\mathfrak{a}$ be organisms. We say $\mathfrak{o}$ affects $\mathfrak{a}$ when there exists a task α in $\mathfrak{s}_{\mathfrak{a}}$ and an input $i \in I_\alpha$ such that completing α from i depends on what $\mathfrak{o}$ does.*

Interpretation. Affect here is not a feeling word. It means one organism can causally change the other organism's future options.

S3 Generalisation optimal learning

S3.1 Maximally uninformative prior implies weakness maximisation

We now give the formal result that justifies w-maxing as a generalisation optimal rule under a maximally uninformative prior. This is the formal backbone of the Monte Carlo benchmark in the main text.

Proposition 1 (Sufficiency of weakness maximisation under a uniform prior)**.** *Let $\mathfrak{v}$ be finite and let $\alpha = \langle I_\alpha, O_\alpha \rangle$ be a task in $L_{\mathfrak{v}}$. Let*

$$U := L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_\alpha)$$

be the set of unseen *statements, meaning statements that never arise as completions of the child inputs. Let ω be a parent task with $\alpha \sqsubseteq \omega$. Assume a maximally uninformative extension model. Conditional on α, the parent selects an additional set $S \subseteq U$ of correct outputs uniformly at random from 2^U, and sets*

$$O_\omega = O_\alpha \cup S.$$

Then for any correct child policy $\pi \in \Pi_\alpha$,

$$\mathbb{P}(\pi \in \Pi_\omega \mid \alpha) = \frac{2^{|\mathrm{Ext}(\pi) \cap U|}}{2^{|U|}}.$$

In particular, among all $\pi \in \Pi_\alpha$, this generalisation probability is maximised by maximising $|\mathrm{Ext}(\pi)|$.

Proof. Fix a correct policy $\pi \in \Pi_\alpha$. By correctness on α,

$$\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi) = O_\alpha.$$

Because $U = L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_\alpha)$, we can decompose

$$\mathrm{Ext}(\pi) = (\mathrm{Ext}(\pi) \cap \mathrm{Ext}(I_\alpha)) \ \cup \ (\mathrm{Ext}(\pi) \cap U) = O_\alpha \ \cup \ (\mathrm{Ext}(\pi) \cap U).$$

Hence, within Π_α, maximising $|\mathrm{Ext}(\pi)|$ is equivalent to maximising $|\mathrm{Ext}(\pi) \cap U|$. Under the stated random extension model, π is also correct for the parent ω precisely when the additional correct outputs $S \subseteq U$ are all permitted by π. That is when $S \subseteq \mathrm{Ext}(\pi) \cap U$. The number of subsets $S \subseteq U$ satisfying $S \subseteq \mathrm{Ext}(\pi) \cap U$ is $2^{|\mathrm{Ext}(\pi) \cap U|}$. The total number of subsets of U is $2^{|U|}$. Therefore

$$\mathbb{P}(\pi \in \Pi_\omega \mid \alpha) = \frac{2^{|\mathrm{Ext}(\pi) \cap U|}}{2^{|U|}}.$$

This quantity is strictly increasing in $|\mathrm{Ext}(\pi) \cap U|$. So it is maximised by maximising $|\mathrm{Ext}(\pi)|$ within Π_α. $\square$

Interpretation. We assume the unknown future cases are chosen uniformly from everything the training task did not rule out. A policy survives the parent task if it already allows those future cases. The more completions a policy allows, the more future cases it covers. So weakness maximisation is Bayes optimal under this ignorance model.

S3.2 Weakness is necessary under the uniform prior

Proposition 2 (Necessity of weakness maximisation under a uniform prior)**.** *In the setting of Proposition 1, let $\pi_w \in \Pi_\alpha$ be a policy that maximises $|\mathrm{Ext}(\pi)|$ among correct child policies. If π is any other correct policy, then*

$$\mathbb{P}(\pi_w \in \Pi_\omega \mid \alpha) \geq \mathbb{P}(\pi \in \Pi_\omega \mid \alpha).$$

If $|\mathrm{Ext}(\pi_w)| > |\mathrm{Ext}(\pi)|$ then the inequality is strict.

Proof. By Proposition 1, for any $\pi \in \Pi_\alpha$,

$$\mathbb{P}(\pi \in \Pi_\omega \mid \alpha) = \frac{2^{|\mathrm{Ext}(\pi) \cap U|}}{2^{|U|}}.$$

This is strictly increasing in $|\mathrm{Ext}(\pi) \cap U|$. Within Π_α, we have $\mathrm{Ext}(\pi) = O_\alpha \cup (\mathrm{Ext}(\pi) \cap U)$, so maximising $|\mathrm{Ext}(\pi)|$ is equivalent to maximising $|\mathrm{Ext}(\pi) \cap U|$. Therefore any policy π_w that maximises $|\mathrm{Ext}(\pi)|$ also maximises the generalisation probability. If $|\mathrm{Ext}(\pi_w)| > |\mathrm{Ext}(\pi)|$, then $|\mathrm{Ext}(\pi_w) \cap U| > |\mathrm{Ext}(\pi) \cap U|$ and the inequality is strict. □

Interpretation. If survival probability increases monotonically with weakness, then choosing the weakest correct policy is not just sufficient. It is necessary.

S3.3 Weakness beyond the uniform prior

The uniform prior from Proposition 1 is the special case where every unseen statement becomes relevant with probability one half, independently. A simple nonuniform generalisation keeps independence but allows different probabilities for different unseen statements.

Theorem 1 (Bayes optimality under independent biased extension priors)**.** *Work in the setting of Proposition 1 but replace the uniform prior on $S \subseteq U$. Assume each $u \in U$ is included in S independently with probability $r_u \in (0, 1)$. Let $A_\pi := \mathrm{Ext}(\pi) \cap U$. Then for any $\pi \in \Pi_\alpha$,*

$$\mathbb{P}(\pi \in \Pi_\omega \mid \alpha) = \prod_{u \in U \setminus A_\pi} (1 - r_u).$$

Equivalently,

$$\log \mathbb{P}(\pi \in \Pi_\omega \mid \alpha) = \sum_{u \in U \setminus A_\pi} \log(1 - r_u) + \textit{constant}.$$

Therefore a Bayes optimal heuristic is obtained by maximising the weighted weakness score

$$W_r(\pi) := \sum_{u \in A_\pi} w_u \qquad \textit{where} \qquad w_u := -\log(1 - r_u).$$

When all r_u are equal, $W_r(\pi)$ is proportional to $|A_\pi|$ and reduces to ordinary w-maxing.

Proof. By Proposition 1, $\pi \in \Pi_\omega$ iff $S \subseteq A_\pi$. Under the independence assumption, $S \subseteq A_\pi$ holds iff every $u \in U \setminus A_\pi$ is excluded from S. Each such exclusion occurs with probability $(1 - r_u)$. Independence gives the product formula. Taking logs yields the second display. Up to an additive constant independent of π, maximising this probability is the same as maximising $\sum_{u \in A_\pi} -\log(1 - r_u)$. □

Interpretation. If some unseen cases are more likely than others, the truly optimal rule is to keep open the cases that are likely to appear. That is a weighted version of weakness. Plain w-maxing is the special case where you really do not know which unseen case will matter.

S3.4 Why description length is not the right proxy

Proposition 3 (Simplicity can be suboptimal)**.** *A shortest policy heuristic based on description length is neither sufficient nor necessary for maximising the generalisation probability in Proposition 1. In particular, there exist tasks for which the shortest correct policy has strictly smaller weakness than another correct policy. Under the uniform prior, the shortest correct policy is then strictly suboptimal.*

Proof. We give an explicit counterexample. Let $\Phi = \{\phi_0, \phi_1, \phi_2, \phi_3, \phi_4\}$. Let the following five sets of programs be the maximal consistent statements.

$$\begin{aligned} m_0 &= \{z, j, k\}, \\ m_1 &= \{j, k, a, b, c, d\}, \\ m_2 &= \{j, k, e, f, g, h\}, \\ m_3 &= \{u_j, j\}, \\ m_4 &= \{u_k, k\}. \end{aligned}$$

Define the vocabulary $\mathfrak{v} := \bigcup_{r=0}^{4} m_r$. Interpret each program $p \in \mathfrak{v}$ as the set of states in which it appears. That is

$$p := \{\phi_r \in \Phi \mid p \in m_r\}.$$

With this construction, a set of programs $\ell \subseteq \mathfrak{v}$ is a statement iff $\ell \subseteq m_r$ for some r. So $L_{\mathfrak{v}}$ is exactly the downward closure of $\{m_0, \ldots, m_4\}$. Now define a $\mathfrak{v}$-task $\alpha = \langle I_\alpha, O_\alpha \rangle$ by

$$I_\alpha = \big\{\{z, j, k\}, \{u_j\}, \{u_k\}\big\}, \qquad O_\alpha = \big\{\{z, j, k\}\big\}.$$

One can check that $\mathrm{Ext}(I_\alpha) = \{\{z, j, k\}, \{u_j\}, \{u_j, j\}, \{u_k\}, \{u_k, k\}\}$.

Consider two candidate policies. Let $\pi_d := \{z\}$ and $\pi_w := \{j, k\}$. Because the only element of $\mathrm{Ext}(I_\alpha)$ that contains z is $\{z, j, k\}$, we have $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi_d) = \{\{z, j, k\}\} = O_\alpha$. So $\pi_d \in \Pi_\alpha$. Similarly, no element of $\mathrm{Ext}(I_\alpha)$ contains both j and k except $\{z, j, k\}$. So $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi_w) = O_\alpha$ and $\pi_w \in \Pi_\alpha$.

The length proxy prefers π_d because $|\pi_d| = 1 < 2 = |\pi_w|$. But weakness prefers π_w. Program z appears only in m_0, which gives $|\mathrm{Ext}(\pi_d)| = 4$. Programs j and k co-occur in m_0, m_1, m_2, which gives $|\mathrm{Ext}(\pi_w)| = 32$. So $|\mathrm{Ext}(\pi_w)| > |\mathrm{Ext}(\pi_d)|$.

Under Proposition 1, generalisation probability increases with $|\mathrm{Ext}(\pi)|$. So the shortest policy π_d is strictly suboptimal. □

Interpretation. You can build vocabularies where the shortest description is a brittle hack. It only works on the training cases. A slightly longer policy can be much weaker in the important sense. It covers more possible future cases.

S4 Full proofs for results used as sketches in the main text

S4.1 Bayesian bound for the measurement failure mode

This section fills in the proof sketch from the main text that links mismatch probability to a likelihood ratio.

Proposition 4 (Likelihood ratio upper bound for decoder mismatch)**.** *Let an evaluator observe a trace y and consider a hypothesis class $\mathcal{H} = \mathcal{H}_{mind} \cup \{h_{noise}\}$ where h_{noise} treats the trace as noise. Assume $\mathbb{P}(h_{noise}) > 0$ and $\mathbb{P}(y \mid h_{noise}) > 0$. If there exists $\varepsilon \in (0, 1)$ such that for every $h \in \mathcal{H}_{mind}$ we have*

$$\mathbb{P}(y \mid h) \leq \varepsilon\, \mathbb{P}(y \mid h_{noise}),$$

then

$$\mathbb{P}(\mathcal{H}_{mind} \mid y) \leq \frac{\varepsilon\, \mathbb{P}(\mathcal{H}_{mind})}{\varepsilon\, \mathbb{P}(\mathcal{H}_{mind}) + \mathbb{P}(h_{noise})}.$$

Proof. By Bayes rule,

$$\mathbb{P}(\mathcal{H}_{mind} \mid y) = \frac{\sum_{h \in \mathcal{H}_{mind}} \mathbb{P}(h)\mathbb{P}(y \mid h)}{\sum_{h \in \mathcal{H}_{mind}} \mathbb{P}(h)\mathbb{P}(y \mid h) + \mathbb{P}(h_{noise})\mathbb{P}(y \mid h_{noise})}.$$

Let $N = \sum_{h \in \mathcal{H}_{mind}} \mathbb{P}(h)\mathbb{P}(y \mid h)$ and $A = \mathbb{P}(h_{noise})\mathbb{P}(y \mid h_{noise})$. Then $\mathbb{P}(\mathcal{H}_{mind} \mid y) = N/(N + A)$. For every $h \in \mathcal{H}_{mind}$ we have $\mathbb{P}(y \mid h) \leq \varepsilon\, \mathbb{P}(y \mid h_{noise})$, so

$$N \leq \sum_{h \in \mathcal{H}_{mind}} \mathbb{P}(h)\, \varepsilon\, \mathbb{P}(y \mid h_{noise}) = \varepsilon\, \mathbb{P}(\mathcal{H}_{mind})\, \mathbb{P}(y \mid h_{noise}).$$

Substitute this into $N/(N + A)$ and cancel the common factor $\mathbb{P}(y \mid h_{noise})$. □

Interpretation. If every mind hypothesis you know how to test fits the data much worse than "it is just noise", Bayes rule forces you to put almost all posterior mass on noise. So a decoding mismatch can make a real mind look invisible.

Proposition 5 (Likelihood ratio bound for mismatch)**.** *Let θ denote the probability that an observation is drawn from the true process rather than a mismatched process. Assume a Beta prior $\theta \sim \mathrm{Beta}(a, b)$. Let D be a dataset of n observations and let k of them match the true process. Then the posterior is $\theta \mid D \sim \mathrm{Beta}(a + k, b + n - k)$. The posterior odds that $\theta > 1/2$ satisfy*

$$\log \frac{\mathbb{P}(\theta > 1/2 \mid D)}{\mathbb{P}(\theta \leq 1/2 \mid D)} \geq (a + k - b - n + k) \log 2.$$

In particular, if k is not substantially larger than $n/2$, the posterior cannot strongly favour $\theta > 1/2$.

Proof. The Beta posterior is standard. For the odds bound, write the posterior density as

$$f(\theta \mid D) \propto \theta^{a+k-1}(1 - \theta)^{b+n-k-1}.$$

Consider the likelihood ratio at θ versus $1 - \theta$. For $\theta \in (1/2, 1)$,

$$\frac{f(\theta \mid D)}{f(1 - \theta \mid D)} = \left(\frac{\theta}{1 - \theta}\right)^{a+k-1-(b+n-k-1)} = \left(\frac{\theta}{1 - \theta}\right)^{a+k-b-n+k}.$$

On $(1/2, 1)$ we have $\theta/(1 - \theta) \geq 1$. So if $a + k - b - n + k \geq 0$, the posterior density is pointwise larger on $(1/2, 1)$ than on $(0, 1/2)$ after reflection, which yields the stated odds lower bound. If $a + k - b - n + k < 0$ the bound is still valid because the right hand side is then negative. □

Interpretation. If you have not seen a strong signal that your decoder matches the world, you cannot justify treating your measurements as ground truth. In that regime the right strategy is to probe.

S4.2 Causal identities are w-max separators

Proposition 6 (Existence of w-maximised causal identities in finite languages)**.** *Assume $\mathfrak{v}$ is finite. If $\mathcal{C}(INT, OBS) \neq \emptyset$, then a w-maximised causal identity exists.*

Proof. When $\mathfrak{v}$ is finite, the language $L_{\mathfrak{v}}$ is finite. So the candidate set $\mathcal{C}(INT, OBS)$ is finite. The set of integers $\{|\mathrm{Ext}(c)| \mid c \in \mathcal{C}(INT, OBS)\}$ is therefore finite and nonempty. So it has a maximum. Any maximiser is a w-maximised causal identity. □

Interpretation. If there are only finitely many signatures to check, the weakest one exists.

Proposition 7 (Finite case w-max implies inclusion minimality)**.** *Assume $\mathfrak{v}$ is finite. Let $c^\star \in \mathcal{C}(INT, OBS)$ be w-maximised. Then there is no other candidate $c \in \mathcal{C}(INT, OBS)$ such that $c \subset c^\star$.*

Proof. Suppose for contradiction that $c \subset c^\star$ for some candidate c. Then every completion of $c^\star$ is also a completion of c. So $\mathrm{Ext}(c^\star) \subseteq \mathrm{Ext}(c)$. Because $\mathfrak{v}$ is finite, $L_{\mathfrak{v}}$ is finite. So strict inclusion implies strict cardinality. Also, c itself is in $\mathrm{Ext}(c)$ but not in $\mathrm{Ext}(c^\star)$. So the inclusion is strict. Hence $|\mathrm{Ext}(c^\star)| < |\mathrm{Ext}(c)|$. That contradicts w-maximality. □

Interpretation. If a weaker separator exists that still works, then the stronger one was never optimal.

S4.3 The first order self is a fixed point of w-maxing

Proposition 8 (Self fixed point)**.** *Let $\mathfrak{o}^1$ be a first order self as in Definition S1.10. Then $\mathfrak{o}^1$ is a w-maximised causal identity for the self intervention discrimination problem. Applying the w-max rule again returns a first order self.*

Proof. By definition, $\mathfrak{o}^1$ is a w-maximised causal identity in the candidate set $\mathcal{C}(INT_{\mathfrak{o}}, OBS_{\mathfrak{o}})$. So it is already a fixed point of the selection rule that chooses w-maximised candidates. □

Interpretation. The first self variable is defined as the weakest self tag that works. So it is literally what you get by w-maxing.

S4.4 Second order selves and Gricean report

This section formalises a simple but sharp dependency claim. If what counts as a correct "report" depends on how an audience interprets a signal, then any signalling policy that is robust across audience variation must represent that audience variation. In the main text, we interpret this representational requirement as a 2ND order self.

Definition 16 (Audience dependent decoding)**.** *Fix an intended meaning set M and a signal set Σ. Let Θ be a set of possible audience epistemic states. For each $\theta \in \Theta$, let $D_\theta : \Sigma \to M$ be the decoding map used by an audience in epistemic state θ.*

Interpretation. Different audiences can decode the same signal differently. We model that by allowing the decoder to depend on an audience state θ.

Definition 17 (Robust Gricean report)**.** *A speaker chooses a signal by a policy π. If the speaker intends meaning $m \in M$ and faces an audience in state $\theta \in \Theta$, the policy outputs a signal $s = \pi(m, \theta)$. We say π is* ***robustly correct*** *if for all $m \in M$ and all $\theta \in \Theta$ we have*

$$D_\theta(\pi(m, \theta)) = m.$$

We say π ***fails to model the audience*** *if $\pi(m, \theta)$ is independent of θ.*

Interpretation. Robust report means the listener recovers the intended meaning across audience variation. Failing to model the audience means you emit the same signal for the same meaning no matter who is listening.

Proposition 9 (Gricean report requires audience modelling)**.** *Assume there exist two audience states $\theta_1, \theta_2 \in \Theta$ and some signal $s \in \Sigma$ such that $D_{\theta_1}(s) \neq D_{\theta_2}(s)$. Then no robustly correct signalling policy can fail to model the audience. Equivalently, any robustly correct policy must condition on a variable that distinguishes θ_1 from θ_2.*

Proof. Suppose for contradiction that π is robustly correct and independent of θ. Fix any meaning $m \in M$. Because π is independent of θ, we have $\pi(m, \theta_1) = \pi(m, \theta_2) =: s$. Robust correctness requires $D_{\theta_1}(s) = m$ and $D_{\theta_2}(s) = m$. So $D_{\theta_1}(s) = D_{\theta_2}(s)$. This contradicts the assumption that some signal has different decodings under θ_1 and θ_2. Therefore any robustly correct policy must depend on θ through some audience modelling variable. □

Interpretation. If two audiences interpret the same signal differently, you cannot talk to both of them correctly using one fixed rule that ignores who they are. You need some representation of the audience.

Corollary 1 (Gricean communication separates 1ST from 2ND order selves)**.** *There exist communication tasks for which no policy can be robustly correct unless it conditions on an audience modelling variable. In the main text's terminology, these are tasks that require a 2ND order self.*

Proof. Pick any setting with Θ and $\{D_\theta\}_{\theta \in \Theta}$ that satisfies the assumption of Proposition 9. By that proposition, robust correctness forces audience modelling. So tasks of this form are solvable only if the agent's embodied language can represent the relevant audience distinctions. □

Interpretation. Some kinds of report are impossible without modelling the audience. That is a clean separation between having a self tag and having a socially usable report.

S4.5 Third order selves and self binding in trust games

This section gives a minimal separation result that motivates 3RD order self structure. It exhibits a multi stage social niche in which even perfect audience modelling is not enough. A publicly legible self binding move is required to make trust rational.

Definition 18 (A two stage trust and commitment game)**.** *Fix a commitment cost c with $0 < c < 1$. Two agents $\mathfrak{a}$ and $\mathfrak{b}$ play the following extensive form game.*

1. *Stage 0. $\mathfrak{a}$ chooses either* Bind B *or* Free F*. Choosing B incurs an immediate cost c to $\mathfrak{a}$ and removes $\mathfrak{a}$'s ability to exploit in Stage 2. Choosing F incurs no cost and leaves Stage 2 unconstrained.*
2. *Stage 1. $\mathfrak{b}$ observes B or F and chooses either* Trust T *or* Do not trust N*. If $\mathfrak{b}$ chooses N, the game ends with payoff $(0, 0)$.*
3. *Stage 2. If $\mathfrak{b}$ chose T then the following happens.*
   - *if $\mathfrak{a}$ chose B in Stage 0, $\mathfrak{a}$ is forced to play* Honour H*.*
   - *if $\mathfrak{a}$ chose F in Stage 0, $\mathfrak{a}$ chooses either* Honour H *or* Exploit E*.*

   Payoffs, before subtracting the binding cost, are

$$u_{\mathfrak{a}}(T, H) = 1, \quad u_{\mathfrak{b}}(T, H) = 1, \qquad u_{\mathfrak{a}}(T, E) = 2, \quad u_{\mathfrak{b}}(T, E) = -1.$$

   If $\mathfrak{a}$ chose B in Stage 0, then c is subtracted from $\mathfrak{a}$'s payoff.

Interpretation. The listener decides whether to trust. If they trust, the actor can either honour or exploit. Binding is a move the actor can make in advance that removes its ability to exploit.

Proposition 10 (A commitment game separates 2ND from 3RD order selves)**.** *In the game of Definition 18.*

1. *If the self binding move B is not available, the unique subgame perfect equilibrium outcome is N with payoff $(0, 0)$.*
2. *If B is available and $0 < c < 1$, there exists a subgame perfect equilibrium in which $\mathfrak{a}$ plays B. $\mathfrak{b}$ plays T after observing B and N after observing F. The Stage 2 outcome is necessarily H, yielding payoff $(1 - c, 1)$.*

Proof. If B is unavailable, then after $\mathfrak{b}$ chooses T the actor $\mathfrak{a}$ chooses between H and E. $\mathfrak{a}$ strictly prefers E because $2 > 1$. Anticipating this, $\mathfrak{b}$ strictly prefers N to T because $0 > -1$. So the unique subgame perfect equilibrium outcome is N with payoff $(0, 0)$. Now allow B. In the subgame following (F, T) the same reasoning applies. $\mathfrak{a}$ plays E, so $\mathfrak{b}$'s best response after observing F is N. In the subgame following $(B, \cdot)$, $\mathfrak{a}$ is forced to play H, so $\mathfrak{b}$'s best response after observing B is T because $1 > 0$. Anticipating these best responses, $\mathfrak{a}$ compares its Stage 0 payoffs. Choosing F yields 0 since $\mathfrak{b}$ chooses N. Choosing B yields $1 - c$. Because $0 < c < 1$, we have $1 - c > 0$ so $\mathfrak{a}$ strictly prefers B. This strategy profile is sequentially rational in every subgame. So it is subgame perfect and yields the cooperative outcome $(1 - c, 1)$. □

Interpretation. This is backward induction. If the actor is unbound, exploitation is optimal and the listener anticipates that and refuses to trust. If the actor binds, exploitation is unavailable so trust becomes rational.

Remark. In the main text we interpret a publicly legible self binding move as a minimal proxy for a 3RD order self operation. It is a policy about policies. Within Stack Theory, the same point can be expressed as a w maxing construction. Among all self binding constraints that make trust viable, w maxing selects a maximally weak one. It is strong enough to rule out exploitative completions, and weak enough to preserve flexibility everywhere else.

S5 Experiments and plots

S5.1 Experiment S1: Weakness beats description length

These experiments are taken from the workshop paper Bennett [2026]. They are primarily illustrative, minimal demonstrations of the theoretical claims.

Goal. We test the claim that weakness, not description length, is the quantity that predicts generalisation under an ignorance prior.

World generation. For each world we sample. A finite environment Φ. A finite vocabulary $\mathfrak{v} \subseteq 2^{\Phi}$. A random child task α and a random parent task ω with $\alpha \sqsubseteq \omega$. The parent is generated by adding extra outputs drawn from a prior over 2^{U}.

Heuristics. A heuristic is a rule that chooses one policy from Π_{α}. We compare. A w-max heuristic that chooses a policy with maximal weakness. A shortest policy heuristic that chooses a policy with minimal description length in the given encoding. A random heuristic that chooses uniformly from Π_{α}.

Regret. Regret is measured in bits. It is the expected negative log probability penalty for generalising to the sampled parent, relative to the Bayes optimal heuristic under the stated prior.

Table S1: Experiment S1 summary. Mean regret in bits across worlds. Lower is better.

Heuristic	Uniform prior	Nonuniform prior
w-max	0.0	0.225
shortest	13.4	7.27
random	70.2	63.5

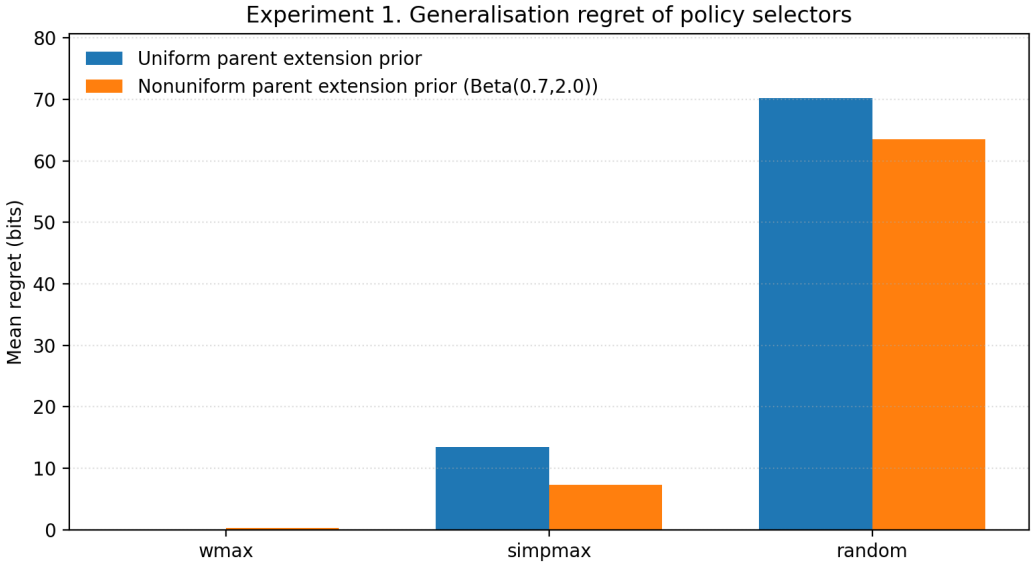

Figure S1: Experiment S1. Regret in bits for three heuristics under two priors. w-max is optimal under the uniform prior by Proposition 1. It remains near optimal under the nonuniform prior.

S5.2 Experiment S2: Probing under decoder mismatch

Goal. We test the claim that under decoder mismatch, passive signalling is fragile while short probing closes most of the gap to an oracle.

Table S2: Experiment S2 summary. Success rates for communication under mismatch.

Method	Stationary decoder	Nonstationary decoder
passive	0.503	0.503
probe 2	0.926	0.928
oracle	0.961	0.961

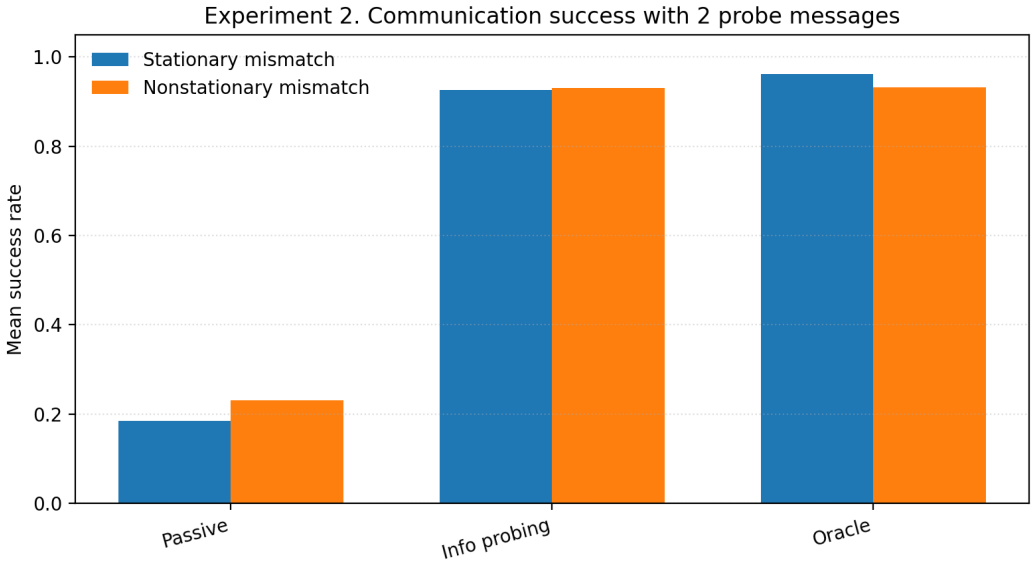

Figure S2: Experiment S2. Passive signalling stays near chance. Two short probes achieve high success in both stationary and nonstationary settings.

S5.3 Experiment S3: Commitment based trust

Goal. We test the claim that credibility requires commitment.

Table S3: Experiment S3 endpoints. Mean trust rates at horizon endpoints for binding and nonbinding commitments.

Commitment type	$H = 1$	$H = 5$
binding	0.442	0.950
nonbinding	0.000	0.000

S6 Law of the stack and a valence first causality principle

S6.1 The Law of the Stack bottleneck

The main text uses The Law of the Stack as a bottleneck result. The Law of the Stack was published in Bennett [2024]. It says higher level expressivity is exponentially limited by how many completions are left

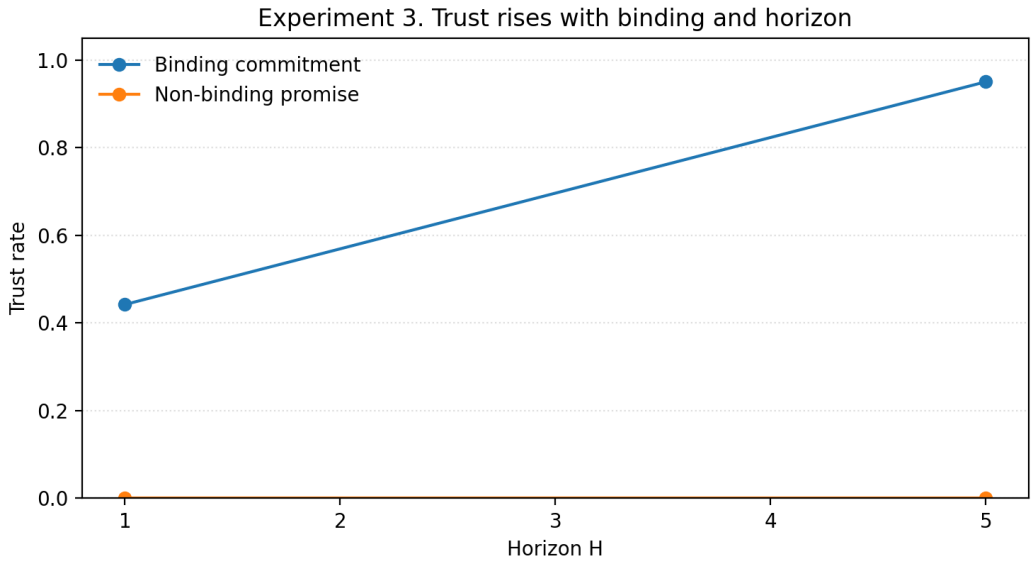

Figure S3: Experiment S3. Trust rates at horizon endpoints. Binding dominates once the interaction has future.

open below.

Theorem 2 (The Law of the Stack)**.** *Consider adjacent layers i and $i+1$ and suppose*

$$\mathfrak{v}^{i+1} = \{\, T(o) \mid o \in \mathrm{Ext}(\pi^i) \,\}.$$

Let γ^{i+1} be the induced layer $i+1$ task. Then

$$\sup_{\pi \in \Pi_{\gamma^{i+1}}} |\mathrm{Ext}(\pi)| \le 2^{|\mathrm{Ext}(\pi^i)|}.$$

If $|\mathrm{Ext}(\pi^i)| < \infty$ then the finite bottleneck form holds

$$\epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}| \le 2^{|\mathrm{Ext}(\pi^i)|}.$$

Proof. By definition,

$$\mathfrak{v}^{i+1} = \{\, T(o) \mid o \in \mathrm{Ext}(\pi^i) \,\}.$$

So $\mathfrak{v}^{i+1}$ is the image of $\mathrm{Ext}(\pi^i)$ under $o \mapsto T(o)$. Therefore $|\mathfrak{v}^{i+1}| \le |\mathrm{Ext}(\pi^i)|$. Every statement in $L_{\mathfrak{v}^{i+1}}$ is a subset of $\mathfrak{v}^{i+1}$. So $L_{\mathfrak{v}^{i+1}} \subseteq 2^{\mathfrak{v}^{i+1}}$. Hence

$$|L_{\mathfrak{v}^{i+1}}| \le 2^{|\mathfrak{v}^{i+1}|} \le 2^{|\mathrm{Ext}(\pi^i)|}.$$

Now fix any $\pi \in \Pi_{\gamma^{i+1}} \subseteq L_{\mathfrak{v}^{i+1}}$. By definition of extension, $\mathrm{Ext}(\pi) \subseteq L_{\mathfrak{v}^{i+1}}$. So

$$|\mathrm{Ext}(\pi)| \le |L_{\mathfrak{v}^{i+1}}| \le 2^{|\mathrm{Ext}(\pi^i)|}.$$

Taking the supremum over $\pi \in \Pi_{\gamma^{i+1}}$ gives the first claim. For the finite bottleneck form, assume $|\mathrm{Ext}(\pi^i)| < \infty$. Then $\mathfrak{v}^{i+1}$ and $L_{\mathfrak{v}^{i+1}}$ are finite. So the supremum in the definition of utility is a maximum. Choose $\pi^\star \in \Pi_{\gamma^{i+1}}$ that attains it. Because $\pi^\star$ is correct, we have $O_{\gamma^{i+1}} \subseteq \mathrm{Ext}(\pi^\star)$. All sets are finite, so

$$|\mathrm{Ext}(\pi^\star)| = |O_{\gamma^{i+1}}| + |\mathrm{Ext}(\pi^\star) \setminus O_{\gamma^{i+1}}| = |O_{\gamma^{i+1}}| + \epsilon(\gamma^{i+1}).$$

Also $|\mathrm{Ext}(\pi^\star)| \le 2^{|\mathrm{Ext}(\pi^i)|}$ by the first part. Rearranging gives the second claim. □

Interpretation. If the lower layer leaves only a small number of consistent refinements open, then the next layer can only name a limited number of distinct things. It is a combinatorial bottleneck.

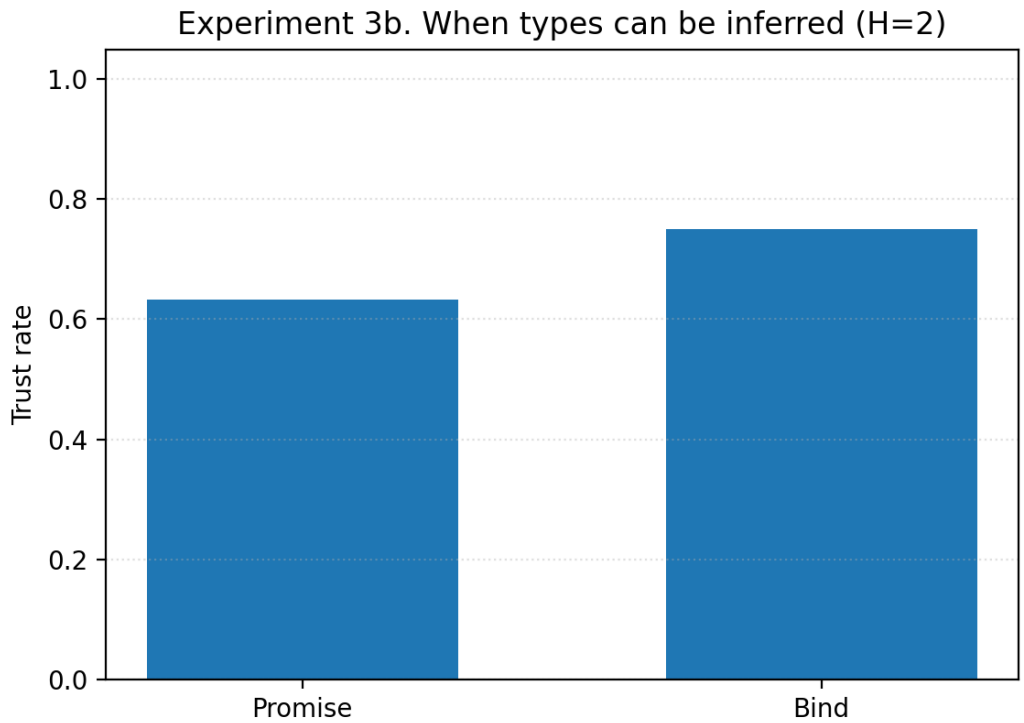

Figure S4: Experiment S3. Distribution of game types in the benchmark sample.

S6.2 Free energy floor from hard commitments

We can connect the bottleneck to a simple free energy proxy.

Corollary 2 (Free energy floor)**.** *Assume the finite setting of Theorem 2. Fix a layer $i+1$ viability statement $\mu \in L_{\mathfrak{v}^{i+1}}$. For any policy $\pi \in L_{\mathfrak{v}^{i+1}}$ define*

$$\Omega_\pi := \mathrm{Ext}(\pi) \cap \mathrm{Ext}(\mu).$$

Assume $\Omega_\pi \neq \emptyset$. Let q_π be the uniform distribution on Ω_π and let p_μ be the uniform distribution on $\mathrm{Ext}(\mu)$. Define the base 2 free energy proxy

$$F_2(q_\pi) := D_{\mathrm{KL},2}(q_\pi \,\|\, p_\mu).$$

Then

$$F_2(q_\pi) = \log_2|\mathrm{Ext}(\mu)| - \log_2|\Omega_\pi|.$$

Also $|\Omega_\pi| \leq 2^{|\mathrm{Ext}(\pi^i)|}$. So

$$F_2(q_\pi) \geq \log_2|\mathrm{Ext}(\mu)| - |\mathrm{Ext}(\pi^i)|.$$

Proof. The KL identity for uniform distributions is direct. If q is uniform on a set of size m and p is uniform on a set of size n containing that set, then $D_{\mathrm{KL},2}(q\|p) = \log_2(n/m)$. Here $n = |\mathrm{Ext}(\mu)|$ and $m = |\Omega_\pi|$.

For the bound, we have $\Omega_\pi \subseteq L_{\mathfrak{v}^{i+1}}$. Also $L_{\mathfrak{v}^{i+1}} \subseteq 2^{\mathfrak{v}^{i+1}}$. So $|\Omega_\pi| \leq 2^{|\mathfrak{v}^{i+1}|}$. By Theorem 2 we have $|\mathfrak{v}^{i+1}| \leq |\mathrm{Ext}(\pi^i)|$. Combine these inequalities. □

Interpretation. If you over constrain the lower layer, the number of viable futures at the next layer is capped. That puts a floor under free energy.

S6.3 A valence first psychophysical causality principle

We can now state the proof idea requested in the main text. Generalisation optimal learning pushes the primitive ontology toward valence. Anything else is a hard commitment that shrinks the viable future set and raises the free energy floor.

Definition 19 (Hard primitive commitments)**.** *Fix a language $L_{\mathfrak{v}}$. Let $C \subseteq L_{\mathfrak{v}}$ be a set of commitments. We say a learning rule has* ***hard primitives*** *C if it restricts the policy space to policies that include every commitment in C. For a task α this restricted class is*

$$\Pi_\alpha^C := \{\pi \in \Pi_\alpha \mid \forall c \in C,\ c \subseteq \pi\}.$$

Interpretation. Hard primitives are constraints you force every candidate policy to contain. They are not learned. They are assumed in advance.

Theorem 3 (The Psychophysical Principle of Causality)**.** *Valence precedes representation under generalisation optimality. Fix a viability statement $\mu \in L_{\mathfrak{v}}$. For any policy $\pi \in L_{\mathfrak{v}}$ define its viable continuation set*

$$\Omega_\pi := \mathrm{Ext}(\pi) \cap \mathrm{Ext}(\mu).$$

Let q_π be the uniform distribution on Ω_π and let p_μ be the uniform distribution on $\mathrm{Ext}(\mu)$. When $\Omega_\pi \neq \emptyset$, define the base 2 free energy proxy

$$F_2(\pi) := D_{\mathrm{KL},2}(q_\pi \,\|\, p_\mu) = \log_2|\mathrm{Ext}(\mu)| - \log_2|\Omega_\pi|.$$

Let $C \subseteq L_{\mathfrak{v}}$ be a nonempty set of hard primitive commitments. Assume C contains at least one commitment c that is not forced by viability. That means there exists some viable continuation $o \in \mathrm{Ext}(\mu)$ with $c \not\subseteq o$.

Let π be any policy with $\Omega_\pi \neq \emptyset$. Assume also that Ω_π contains at least one viable continuation that violates c. Then either $\Omega_{\pi\cup c} = \emptyset$ or

$$F_2(\pi \cup c) > F_2(\pi).$$

So adding an unnecessary primitive commitment strictly raises free energy whenever it rules out even one viable future. In particular, any learning rule that enforces non valence representational primitives as hard coded commitments pays a strict free energy penalty on tasks where the evidence does not already force those commitments. So under a generalisation optimality objective, valence is the only safe primitive. Everything else must be learned as a revisable causal identity.

Proof. Fix c as in the statement. We have $\mathrm{Ext}(\pi \cup c) \subseteq \mathrm{Ext}(\pi)$ because every completion of $\pi \cup c$ is a completion of π. Intersect with viability to get $\Omega_{\pi\cup c} \subseteq \Omega_\pi$.

If $\Omega_{\pi\cup c} = \emptyset$ we are done. Otherwise, by assumption there exists $o \in \Omega_\pi$ with $c \not\subseteq o$. Then $o \notin \Omega_{\pi\cup c}$. So the inclusion is strict and $|\Omega_{\pi\cup c}| < |\Omega_\pi|$.

Apply the identity $F_2(\pi) = \log_2|\mathrm{Ext}(\mu)| - \log_2|\Omega_\pi|$. Since the numerator is fixed and the denominator shrinks, $F_2(\pi \cup c) > F_2(\pi)$. □

Interpretation. If you hard code extra ontology, you eliminate viable futures. That puts a floor under free energy and it blocks generalisation optimality. So the only thing you are allowed to treat as primitive is the viability signal itself. That is what we call valence. Everything else has to be learned as a revisable causal identity.

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